ab18gu@gmail.com • linkedin.com/in/abgup • github.com/ab12gu
Portfolio: **abgup.com** & **notes.abgup.com**
Current: Seattle, WA 98104
U.S. Citizen

Abhay Gupta

After taking leave to help with family, I have been able to sustainably build a product design firm and teach simultaneously. I am willing to transition back into a corporate role if the right job arises!

SKILLS

Languages	Frameworks	Software Tools	Modeling/Analysis	Hardware Tools
Python	WinForms/WPF	Git	SolidWorks	3D Printers
C# & Lua	Flask	Cmake	Siemens NX	Lathe/Mill
Javascript & VBA	Jekyll	Jira/Azure	OnShape/AutoCAD	Drill Press
HTML/CSS	Cordova	Docker	Ansys CFX	Bandsaws
Bash & Powershell	JQuery	Figma	Star CCM+	Sanders/Grinders
Vimscript	Next.js	Microsoft Office	Abaqus	Oscilloscopes
C/C++	LaTeX & Maple	Linux	FreeCAD	Soldering
LabView	Matlab/Simulink	Android/iOS	Fusion 360	Function Generators

EDUCATION

University of Washington — M.S. Robotics & Data Science — Sep 2018 - Dec 2019
- Thesis - 3D Graphics & Regression Models || 3.6
- Advisors: Steve Brunton & Ashis Banerjee

Python || MATLAB || C++ — OpenGL || ROS

Santa Clara University — M.S. Computer Engineering — Sep 2017 - Jun 2018
- Half Completed || Transferred to University of Washington || 3.8

B.S. Mechanical Engineering — Sep 2014 - Sep 2017
- Entrepreneurship minor || Graduated in 3 years with honors || 3.6

MATLAB || LaTeX || C || Simulink || LabVIEW || Maple — SolidWorks || Abaqus || Star-CCM+
Lathe || Mill — Oscilloscopes

INDUSTRY EXPERIENCE

Pebble Products LLC
Founder — Aug 2022 - Present
- Develop/prototytpe new hardware/software products
- Contract engineering services to local companies/individuals

notes.abgup.com — pebbleproducts.com

Bishop Blanchet High School
CS & Robotics Teacher — Dec 2020 - Present
- Teach high school students engineering skills to design/build a autonomous robot
- Teach computer science in Python for Rainier Scholars

Java || Python — OnShape

Kawasaki Robotics
SW Engineer, Robotics — Jun 2022 - Aug 2022
- Developed software to move wafer handling (scara) robots
- Tested physical robotics arms on local and vendor facilities to ensure functionality
- Optimized for throughput and reach requirements to maximize computer chip production capability

AS - Domain Specific Language

SummerBio
SW Engineer, Robotics — Oct 2021 - Feb 2022
- Developed software drivers (6 DOF arms, benchtop systems) for VWorks
- Built communication platforms through both Ethernet and serial port protocols
- Built hardware testing rigs to ensure new hardware/drivers work with automation line
- Unit tested drivers through manual isolation testing and automated via c# test framework, xUnit

C# || Figma — OnShape

Lam Research
SW/ME Engineer || 1y6m — May 2020 - Oct 2021
- Led and assisted Android app development via Cordova (JS) framework
- Developed/Maintained software to optimize a 6 DOF robotic arm via C# WinForms
- Built communication protocols for hardware components via Modbus, Ethernet, and Bluetooth protocols
- Developed HMI software and tested HW/SW in a clean room environment via systematic approach
- Analyzed test results via data science techniques to ensure repeatability and reliability metrics
- Led development of and maintained a fishbone diagram issue diagnosis application with offshore developers via C# Xiamarin Framework on Windows and Android
- Automated Android content upload and verification via VBA excel automation tools
- Worked alongside machinists to develop custom test rigs and hardware components (sheet metal, aluminum/steel parts, injection molded designs, etc) for robotics arm applications
- Designed and printed custom 3D components on a MakerBot and local venders
- Designed a custom electronics cart via Siemens NX fitting custom components and facility constraints

C# || JavaScript || Python || VBA || Figma — NX || Teamcenter

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CSAA Insurance
Physics Consultant || 2y

- Evaluated novel engineering and physics aspects for patent applications
- Researched alternative approaches for products and methods

Oct 2018 - Oct 2020

INDUSTRY EXPERIENCE (continued...)

University of Washington
Research/Teach Asst. || 1y3m

- Taught and led students in graduate mathematics courses for engineering
- Developed a custom physics simulation of Optical Tweezers via OpenGL in C++
- Created a digital twin of a compliant motor and optimized model via novel regression techniques.

C++ || Python   OpenGL || ROS

Sep 2018 - Dec 2019

Santa Clara University
Research/Teach Asst. || 1y9m

- Simulated biological membrane fracture through AI methods (*see paper*)
- Assisted teaching course in Numerical Analysis to undergraduate engineering students

Matlab/LaTex

Sep 2016 - Jun 2018

INTERNSHIPS || 2y

Microvision
SDE Intern || 3m

- Modeled the response of Lidar activated SiPM (Silicon photomultipliers)

Simulink || MATLAB   LTSpice

Summer 2019

TheraNova
SDE Intern || 3m

- Developed a python software analysis system to understand gait measurements
- Ensured the software analysis is accurate for 80+ patients

Python || MATLAB

Summer 2018

Valeo
Systems Engr Intern || 3m

- Produced hardware and software demos for automotive OEMs
- Collaborated with start-ups and OEMs to develop new cabin safety features

VBA

Spring 2018

Pentair
ME/EE Intern || 7m

- Optimized performance of steady state and transient phases of circuit breakers
- Laboratory tested and simulated rail heating to melt snow

Ansys CFX || Solidworks   Oscilloscopes, Function Generators, DC/AC Power Supplies, Soldering

Winter 2017

Accel Biotech
ME Intern || 3m

- Prototyped medical device components and test assemblies
- Supported mechanical, electrical, and software design of a blood diagnostic device

SolidWorks   Oscilloscopes

Summer 2016

Caltrans
ME Intern || 3m

- Reviewed and advised on structural testing for next generation locomotives
- Designed a floor plan using Microsoft Visio & participated in vendor meetings

Summer 2015

PUBLICATIONS, CONFERENCES & PATENTS

Automobile Damage Detection Using Thermal Conductivity — Dec 2018
J. Schow, and A. Gupta (US Patent)

A Cellular Automaton for Modeling Non-Trivial Biomembrane Ruptures — July 2018
A Gupta, G. Reint, I. Gozen, and M. Taylor
Presented at *The 13th World Congress in Computational Mechanics (WCCM)* — Sep 2018
Session: Novel Mathematical Models and Computational Methods, New York, NY
Published in *Soft Matter*

Certifications:
Engineer-In-Training, State of CA — May 2018
Solidworks CSWA — Mar 2015

VOLUNTEERING ACTIVITIES

Auto Angels, Bellevue Car Mechanic — Jun 2024 - Present
Rainier Scholars, Seattle Computer Science Curriculum TA/Development Assistant Python — Jun 2024 - Present
The Bikery, Seattle, Bicycle Board Member & Technician — Nov 2023 - Present
Capitol Hill Tool Library, Seattle Shop Manager Onshape, Machine Shop Tools (metal/wood/acrylic) — May 2024 - Present
San Jose Bicycle Coalition (SJBC), Software & Data Consultant — Jul 2020 - Present
Santa Clara University, Alumni Engineering Mentor HTML, CSS, Javascript, Python — Oct 2023 - Present
San Jose Bicycle Clinic, Bicycle Technician — Jul 2020 - Present
FIRST Robotics Competition (FRC) 254, 5940, 1983, & 4180, Engineering Mentor JAVA, Onshape, SolidWorks — Aug 2021 - Present
Reddit Group: r/ControlTheory, Moderator — Dec 2018 - Present

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SOFTWARE PROJECTS

206 Bike Polo
Web Developer
- Update and maintain website using Javascript framework (Next.js)

HTML || CSS || Javascript | Next.js | 206bikepolo.com

San Jose Bicycle Coalition
Software/Data Consultant
- Use OCR package to automate uploading of sign in sheets into salesforce

Python | Salesforce | bikesiliconvalley.org

Find a Paint
Web Developer
- Developed website to compare paint brands to help high school mom find corresponding paint for work

Python || HTML/CSS || JS | Flask | findapaint.com

Masala Blend
Web Developer
- Help stepmother build webpage to sell homemade spices locally

HTML || CSS || Javascript | Shopify || Jekyll | masalablend.com

Personal Website & Resume
Web Developer
- Build personal website via github pages & programmatically render resume via Latex

HTML || CSS || JS || Latex | Jekyll || Xetex | abgup.com

Peer Porfolio Consulting
Web Developer
- Assist undergraduates build their resume & web porfolio to gain industry experience

HTML || CSS || JS || Python | Latex || Django | github.com/carly85/personal-resume

Bike Components Database
Web Developer
- Build webpage & database to provide comparision between bicycle components

Markdown | github.com/ab12gu/freewheels

HARDWARE PROJECTS

Cap Hill Tool Library
3D Printing Lead
- Assist community members and create models/prints for in-house repairs and personal projects

OnShape

Bike Polo
Mallet Design/Fabrication
- Model, 3D print, and fabricate custom bike polo mallet
- Iterate through design process, optimizing for weight and reduction of stress concentration/shattering

Onshape | Drill, Drill Press, Bandsaw

Custom LEDs
Smart LED Lighting
- Build custom LED timed lighting to match the circadium rhythm of the sun
- Change lighting within building from blue light to gradually hit red light throughout day

C/C++ | Rasberry Pi

Vehicle Builds and Repair
Repair Tech
- Build custom bicycles in community to encourage human powered tranportation
- Repair small and large vehicles in community (Electric Unicycles, Motored vehicles, etc)

Torque wrenches, hydraulic lifts, etc | Soldering iron, oscilliscopes, etc

Bike Component Automation
Automation Lead
- Design and build wireless shifting and cam shafts on bicycles with peers

FreeCAD || C/C++ | Arduino Uno | github.com/ab12gu/bicycle-projects

HOBBIES

Sports	Sports	Indoor	Indoor	Miscellaneous
• Juggling	• Walking	• Board Games	• Weight Lifting	• Coding
• Handstands	• Tennis	• Rubix Cube Solving	• Reading	• Blogging
• Bicycling	• Calesthetics	• Dancing	• Video Games	• Content Creation
• Unicycling	• Running	• Music & Movies	• Cooking	• Comedy
• Bike Polo	• Skateboarding	• Drawing	• Yoga	• Reddit
• Basketball	• Rollerskating	• Painting	• Socializing	• Tinkering

For more: **abgup.com**